Ayush Sawant

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Skills

Languages: C++, JavaScript, TypeScript, SQL

Frontend: React.js, Next.js, Tailwind CSS

Backend: Node.js, Express.js, Socket.IO, REST APIs

Databases & Messaging: PostgreSQL, MongoDB, Prisma, Drizzle ORM, Apache Kafka, KafkaJS

Tools & Concepts: Git, Docker, Postman, Vercel, Supabase, Data Structures and Algorithm, Object Oriented Programming, Database Design, Event-Driven Architecture

Education

Dr. D. Y. Patil Institute of Technology, Pimpri, Pune Aug 2024 - Jun 2028
B.E. in Artificial Intelligence and Data Science ***CGPA: 8.77/10***
Relevant Coursework: Object Oriented Programming, Databases, Discrete Maths, Data Structures and Algorithms, Operating Systems, Advance Data Structures and Algorithms, Artificial Intelligence, Data Science

Project Work

- **Atten-Dance** [Live Demo] *React, TypeScript, Express.js, PostgreSQL, Drizzle ORM, Better Auth*
  - Built and deployed an attendance tracker used by **15+** students, with **1100+** records, enabling students to track attendance, calculate skip limits, and plan future attendance (Express API on Vercel serverless, Supabase PostgreSQL)
  - Designed a normalized schema for semesters, subjects, and attendance; implemented session-based auth with email/password and Google OAuth
  - Built safe-skip and recovery calculators, including a combination solver that finds every (lectures, labs) pair that lifts the weighted average above the threshold
  - Added a what-if predictor with live current-vs-predicted diffs, Recharts analytics, and one-click semester sharing via 6-character codes
- **Polloye** [Live Demo] *Next.js, TypeScript, PostgreSQL, Prisma, Socket.IO, Docker, Azure*
  - Built a Kahoot-style live quiz platform (join by 6-digit code or QR) with MCQ, multi-select, and true/false questions, speed-weighted scoring, and leaderboards
  - Split the system into a stateless Next.js CRUD/auth server and a stateful Socket.IO server, linked by short-lived shared JWTs for authenticated WebSocket handshakes
  - Kept live session state in memory and persisted scores only at answer reveal, keeping DB writes off the hot path; first submission per question wins
  - Designed role-specific reconnect handling: quizzes auto-advance if the host drops, and rejoining participants can't answer a question that went live while they were away
  - Load-tested with **80** concurrent participants;
- **Axon** [npm] *TypeScript, OpenAI, Anthropic, Gemini APIs*
  - Built and published `axon-llmrouter`, a TypeScript SDK that routes each prompt to a Frontier, Balanced, or Fast model tier across OpenAI, Anthropic, Gemini, and any OpenAI-compatible endpoint
  - Designed a two-stage router: a heuristic gate sends trivial prompts straight to the Fast tier, and the rest go to an LLM judge that scores four axes, mapped to a tier by lookup table
  - Implemented tier fallback with irreversibility-aware failure handling: if a high-stakes request fails on Frontier, inference halts and the host app decides, instead of silently downgrading
  - Built an eval harness over a hand-labeled CSV reporting tier-match rate, gate vs. judge accuracy, and per-axis agreement; **68%** tier match on **219** prompts

Leadership & Activities

- **Social Media Manager – Zion Technical Fest, DY Patil Institute of Technology:** Created promotional content and managed social media posts for the college technical festival. Produced short-form video content that reached more than 300,000 views, helping increase visibility and engagement for the event.
- **Volunteer – My Prakriti Community Initiative:** Regularly participate in weekly plogging drives focused on environmental cleanliness by collecting waste while jogging in public spaces, contributing to community awareness about sustainable practices.